Lecture 3: Regularization and Model Stability

ECON 417 Business Forecasting

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How Lecture 3 Fits the Course

- Lecture 2 taught us to distrust good-looking fit without honest validation.
- Lecture 3 asks a new question: **even if accuracy is acceptable, is the model stable?**
- Lecture 4 will extend this logic to more flexible nonlinear models.

Key message: A forecasting model should not change its story every time the sample window moves a little.

Agenda (Today)

1. Why Stability Matters
2. Multicollinearity and Regularization
3. Implementation Workflow

Coefficient Stability: The New Criterion

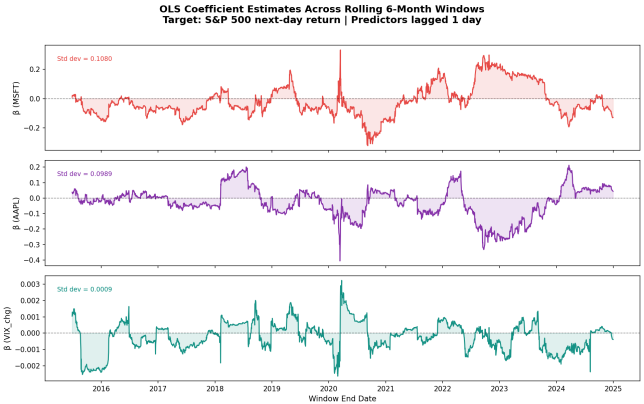
Definition

Coefficient stability asks whether estimated signs and magnitudes stay reasonably similar across nearby training windows.

- Accuracy tells us whether forecasts are close.
- Stability tells us whether the model logic is trustworthy.
- In business settings, both matter.

Key message: A model that flips from "buy" to "sell" every month is hard to trust.

Notebook Evidence: Same Model, Different Story



- The target and predictors stay the same.
- Only the estimation window changes.
- Yet OLS coefficients move sharply.

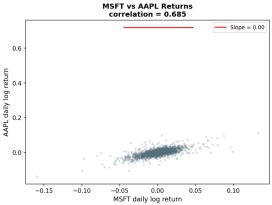
Multicollinearity Creates Fragile Attribution

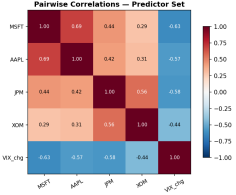
- When predictors overlap heavily, OLS struggles to assign separate credit.
- Small noise changes can create large coefficient swings.
- This is why multicollinearity is a forecasting problem, not just a textbook term.

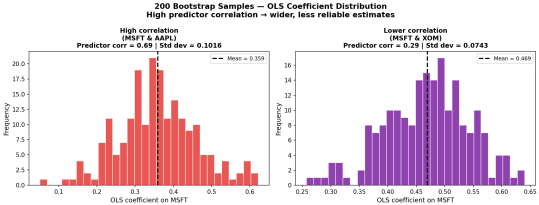
$$\hat{\beta}_{\text{OLS}} = (X^\top X)^{-1} X^\top y$$

Key message: If the predictor matrix is nearly singular, noise gets amplified.

Notebook Evidence: Overlap and Volatility







- Left: predictors share information.
- Right: OLS coefficients become unstable across samples.

Regularization: Why We Shrink Coefficients

Regularization trades a little bias for a lot more stability

OLS
low bias
high variance

→

Ridge
smooth shrinkage

→

Lasso
shrink + selection

Training fit	Coefficient stability	Interpretability
very high	improves	sparser story

concept adapted from: IBM regularization

Ridge Regression

$$\hat{\beta}_{\text{ridge}} = \arg\min_{\beta} \sum_{i=1}^{n} (y_i - x_i^\top \beta)^2 + \lambda \sum_{j=1}^{p} \beta_j^2$$

- Ridge adds an L_2 penalty.
- All predictors stay in the model.
- Coefficients shrink smoothly toward zero.

Interpretation

Ridge is often preferred when many predictors are correlated and all may contain some useful signal.

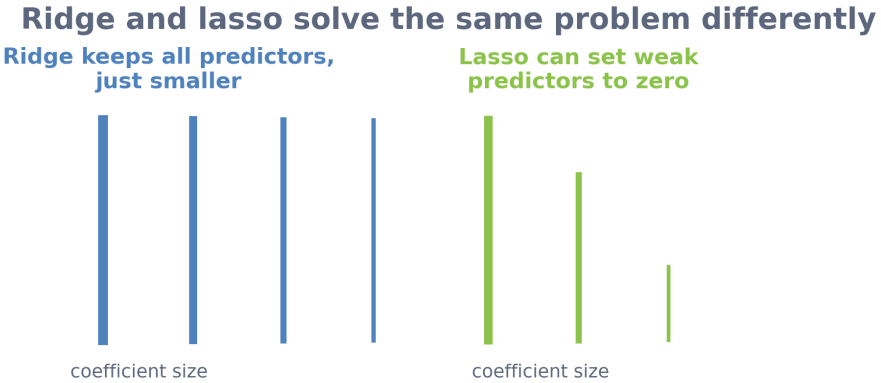
Lasso Regression

$$\hat{\beta}_{\text{lasso}} = \arg\min_{\beta} \sum_{i=1}^{n} (y_i - x_i^\top \beta)^2 + \lambda \sum_{j=1}^{p} |\beta_j|$$

- Lasso adds an L_1 penalty.
- Some coefficients can become exactly zero.
- This creates automatic variable selection.

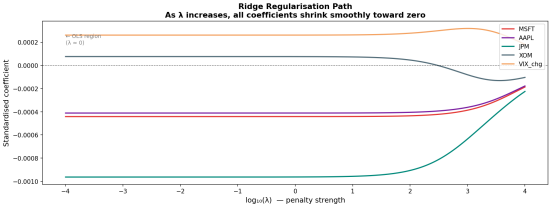
Key message: Lasso gives a shorter story, but its selected variables can change across windows.

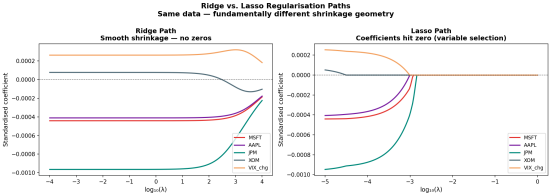
Ridge vs Lasso: Same Goal, Different Mechanism



concept adapted from: IBM ridge regression

Notebook Evidence: Coefficient Paths





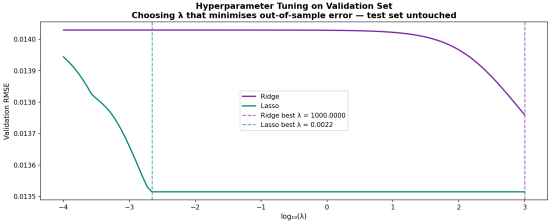
- Ridge compresses coefficients gradually.
- Lasso can push weak predictors all the way to zero.

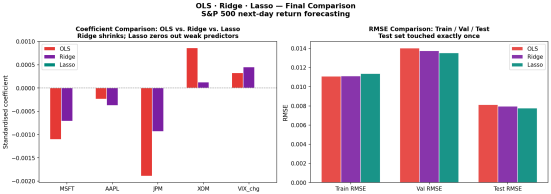
Decision Rule: When to Prefer Which One?

	Ridge	**Lasso**
Main strength	Stability	Sparsity
Best when	Predictors overlap a lot	Many weak features exist
Risk	Harder to simplify	Selection can flip

Key message: Choose by forecasting objective, not algorithm popularity.

How We Compare OLS, Ridge, and Lasso





- Train on the first sample.
- Tune λ on validation data only.
- Compare final models on the untouched test set.

Takeaways

1. Validation measures whether a model generalizes.
2. Regularization improves robustness when OLS is unstable.
3. Ridge is usually the safer first choice in weak-signal forecasting.
4. Lasso is useful when feature screening is a core goal.

Key message: Lecture 4 will ask what happens when we add even more flexibility through nonlinear models.